

Ahmed Body CFD Analysis

Mesh Independence Study and Experimental Validation
25° Slant Angle, $k\text{-}\omega$ SST

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1 Introduction

The Ahmed body [1] is a simplified bluff-body geometry that has become the standard benchmark for automotive aerodynamics CFD validation. Its geometry is fully parametric and its flow field—particularly the slant-angle-dependent wake structure—is extensively documented in the open literature.

This study uses the **25° slant angle** configuration at $U_\infty = 40$ m/s ($Re_L = 2.78 \times 10^6$ on the body length, equivalently $Re_H = 7.68 \times 10^5$ on the body height). Reynolds numbers throughout this report use the kinematic viscosity adopted by the reference experiment, $\nu = 15 \times 10^{-6}$ m²/s [5], which reproduces the Re_H quoted there exactly. Two distinct experimental sources are used, and it is important not to conflate them:

- **Force coefficients** are taken from Meile et al. [4], who measured $C_d = 0.29883 \pm 0.5\%$ on the original geometry at 40 m/s using a six-component force balance, and Meile et al. [3], who report $C_l = 0.345$.
- **Wake velocity profiles** are taken from Lienhart & Becker [2], who published LDA measurements in the wake. That work does *not* report force coefficients, and is therefore not a valid citation for C_d or C_l .

1.1 Objectives and Scope

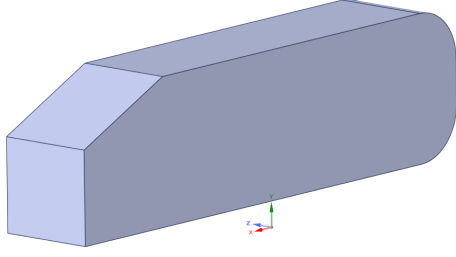
1. Perform a three-level mesh independence study using the GCI methodology [7].
2. Validate the fine-mesh C_d and C_l against the experimental references, ensuring the comparison is like-for-like in both geometry (support stilts) and boundary conditions (ground motion).

The scope is deliberately limited to the **integrated force coefficients**. Validation of the resolved wake structure against the LDA profiles is a separate claim and is not made here; the experimental dataset is included in the repository for subsequent work, and the reasons are set out in Sec. 7.5.

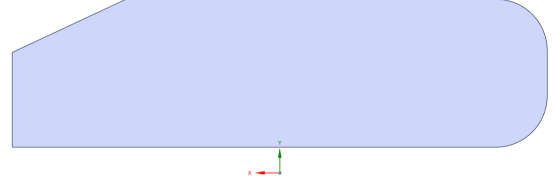
2 Geometry and Computational Domain

2.1 Ahmed Body

Standard dimensions: $L = 1044$ mm, $H = 288$ mm, $W = 389$ mm, ground clearance $g = 50$ mm, slant angle $\alpha = 25^\circ$. A half-model is used with the symmetry plane at $z = 0$.



(a) Isometric view — SpaceClaim

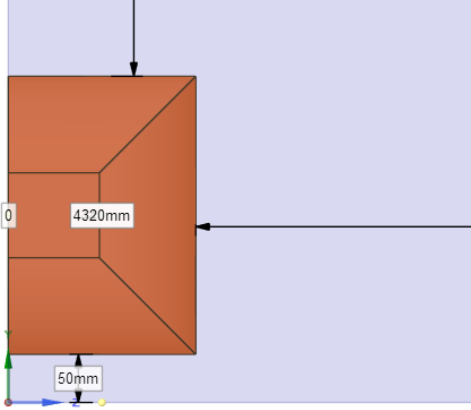


(b) Side view

Figure 1: Ahmed body half-model geometry (25° slant).

2.2 Computational Domain

Domain dimensions follow best-practice guidelines for bluff-body external aerodynamics: upstream $5H$, downstream $15H$, lateral $5H$, top $5H$.



(a) Rear view — 4320 mm downstream, 50 mm ground clearance



(b) Overall domain

Figure 2: Computational domain in SpaceClaim.

2.3 Boundary Conditions

Table 1: Boundary conditions.

Surface	Type	Value
Inlet	Velocity inlet	$U_\infty = 40 \text{ m/s}$, $TI = 0.2\%$
Outlet	Pressure outlet	0 Pa gauge
Ground	Stationary wall	$U = 0$ (final configuration) [†]
Side / Top	Symmetry	—
Symmetry plane ($z = 0$)	Symmetry	—
Body surface	No-slip wall	$U = 0$

[†]The ground was initially specified as a moving wall at $U = 40$ m/s, emulating a rolling road. Both reference experiments in fact ran over a stationary floor; the condition was corrected and the fine mesh rerun. The mesh independence study (Sec. 5) was carried out in the original moving-ground configuration. See Sec. 7.3.

3 Mesh Strategy

3.1 Workflow

Meshes were generated in ANSYS Fluent Meshing 2024 using the *Watertight Geometry* workflow with a **poly-hexcore** volume fill.

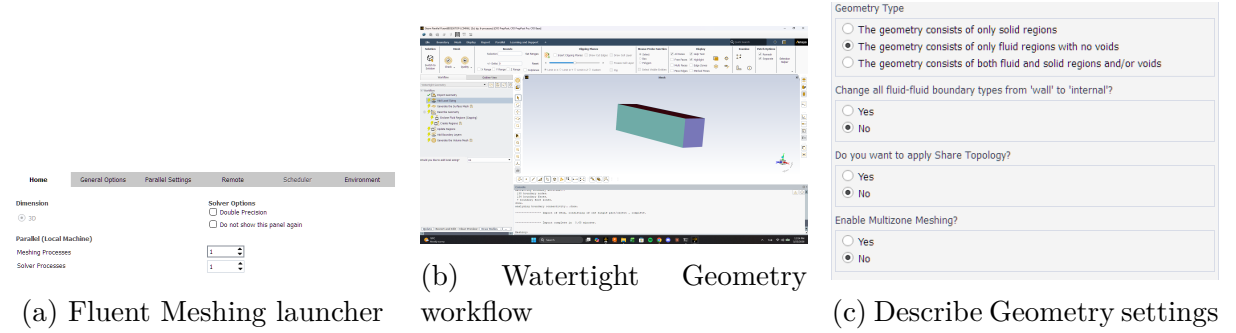


Figure 3: Fluent Meshing setup.

3.2 Boundary Conditions and Region Types

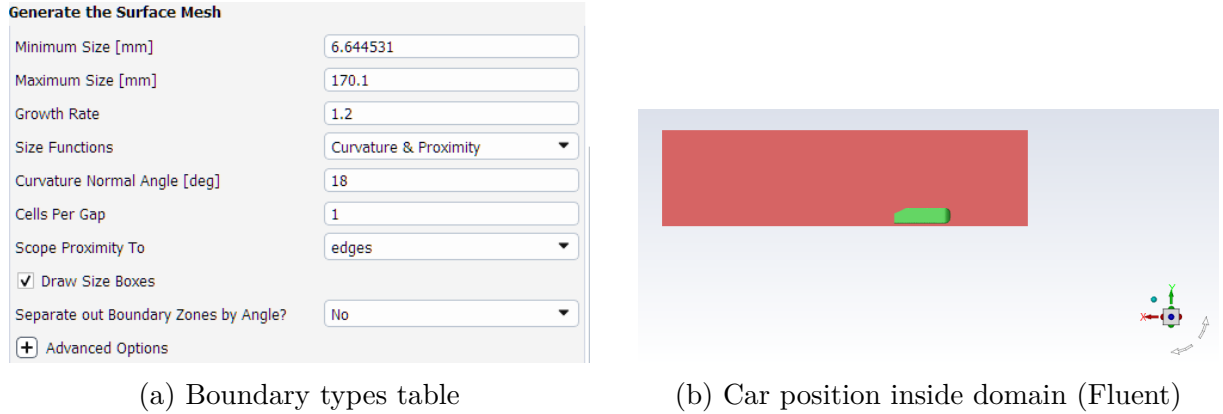


Figure 4: Named selections and boundary assignments.

3.3 Boundary Layer

Wall functions ($y^+ \approx 15\text{--}30$) with aspect-ratio sizing: 8 layers, first aspect ratio 40, growth rate 1.2.

The achieved y^+ range sits in the buffer layer ($5 < y^+ < 30$) rather than clear of it. Neither near-wall strategy is strictly valid there: wall functions assume the log-law ($y^+ > 30$), while low-Reynolds resolution requires $y^+ < 1$. This is a consequence of sizing the first layer by aspect ratio rather than by absolute thickness, and its effects are taken

up in Sec. 7.1 and Sec. 7.4. `scripts/y_plus_calculator.py` reports the first-cell height for both strategies.

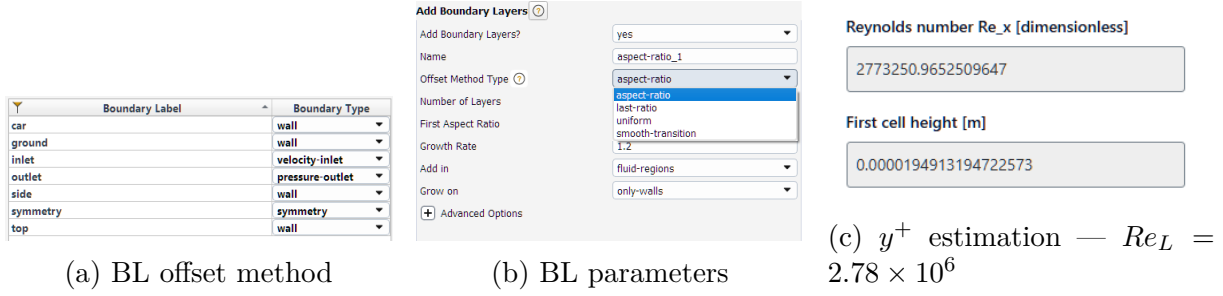


Figure 5: Boundary layer settings and y^+ calculation.

3.4 Volume Mesh and Surface Quality

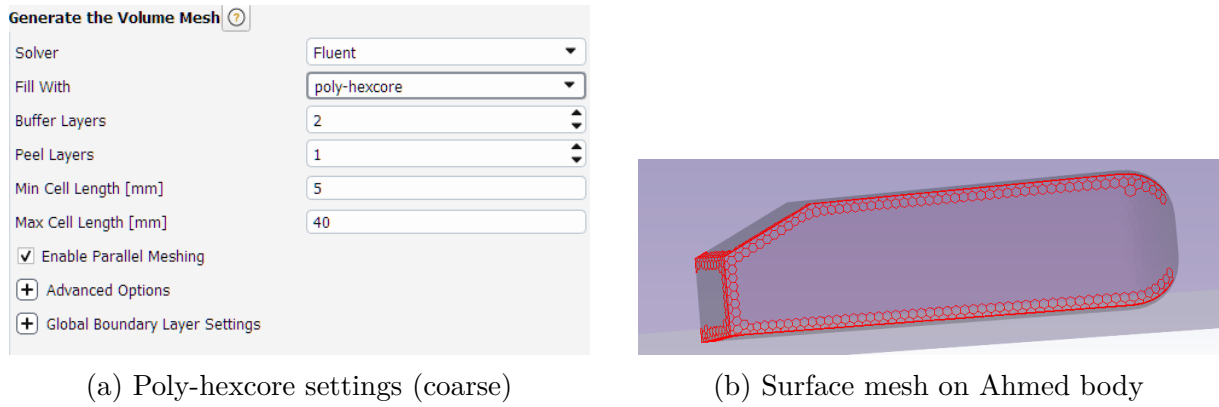


Figure 6: Volume mesh generation. Min cell = 5 mm, Max cell = 40/20/12 mm for coarse/medium/fine.

----- Warning----- The Max Cell Length is updated to reflect the actual max size in the octree hexcore.

Figure 7: Poly-hexcore octree constraint warning during fine mesh generation — max cell size adjusted to next valid power-of-2 multiple of min cell size.

3.5 Mesh Quality Metrics

Volume mesh quality as reported by Fluent's `/mesh/quality` utility:

Table 2: Mesh quality metrics. Orthogonal quality is bounded above by 1 (higher is better); Fluent requires > 0.01 and regards > 0.1 as good. High aspect ratios are expected within prism layers.

Metric	Coarse (370k)	Medium (837k)	Fine (4.4M)
Min. orthogonal quality	0.1638	0.1637	0.1537
Max. tangent skewness*	6.022	6.025	6.427
Max. aspect ratio	122.21	122.72	122.21

*Fluent’s tangent-skewness measure, as reported by `/mesh/quality`. This is not the conventional skewness metric bounded on $[0, 1]$ and should not be read against the usual < 0.95 guideline.

The maximum aspect ratio is effectively identical at all three levels, despite a twelvefold increase in cell count. This is a direct measurement of how the mesh family was constructed: the boundary-layer specification (8 layers, first aspect ratio 40, growth rate 1.2) was held fixed at every level, so refinement acted on the hexcore freestream cells alone while the prism stack remained unchanged.

This bears directly on the mesh independence study. Section 7.4 attributes the apparent order $p = 3.53$ — in excess of the scheme’s formal second-order accuracy — to a mesh family that is not self-similar, with prism and freestream cells refining at different rates and y^+ consequently not held constant across levels. Table 2 supplies the quantitative evidence for that argument, which had until now rested on reasoning alone. Specifying a fixed first-layer *thickness* rather than a fixed aspect ratio would refine both regions together and yield a cleaner family.

Location of the worst cells. All three meshes place them at the same position, all within zone 308: (0.506, 0.0499, 0.178) for coarse, (0.522, 0.0586, 0.186) for medium and (0.518, 0.0500, 0.190) for fine. Interpreted against the geometry, $y \approx 0.050$ m is precisely the ground clearance, $x \approx 0.51$ m is mid-body ($L = 1.044$ m) and $z \approx 0.18$ m lies near the side edge (half-width 0.194 m). The worst cells therefore sit at the *lower side edge of the underbody*, where the prism stack is compressed into a sharp dihedral corner. In each mesh the worst orthogonal-quality cell is simultaneously the worst-skewness cell, with the worst-aspect-ratio cell less than a millimetre away. Refinement does not relocate them, the constraint being geometric rather than resolution-driven. Given $GCI_{\text{fine}} = 0.77\%$, their effect on the solution is not measurable.

A maximum aspect ratio of ≈ 122 against a target first-layer aspect ratio of 40 is expected: the target governs the first layer only, whereas the reported maximum is taken over the entire boundary-layer stack.

4 Solver Settings

Table 3: ANSYS Fluent solver configuration (identical for all mesh levels).

Parameter	Setting
Solver	Pressure-based, steady-state
Turbulence model	$k\text{-}\omega$ SST (Menter 1994 [9])
Pressure-velocity coupling	Coupled
Spatial discretisation	Second-order upwind
Convergence — residuals	$< 10^{-5}$
Convergence — force monitors	$\Delta C_d < 0.001$ over last 200 iterations

5 Mesh Independence Study

5.1 GCI Methodology

For three meshes with cell counts $N_1 > N_2 > N_3$ and quantities f_1, f_2, f_3 , the grid refinement ratios and the differences between successive solutions are

$$r_{21} = \left(\frac{N_1}{N_2}\right)^{1/3}, \quad r_{32} = \left(\frac{N_2}{N_3}\right)^{1/3}, \quad \varepsilon_{21} = f_2 - f_1, \quad \varepsilon_{32} = f_3 - f_2. \quad (1)$$

The apparent order of accuracy p follows from Celik et al. [7]:

$$p = \frac{1}{\ln r_{21}} \left| \ln \left| \frac{\varepsilon_{32}}{\varepsilon_{21}} \right| + q(p) \right|, \quad (2)$$

$$q(p) = \ln \left(\frac{r_{21}^p - s}{r_{32}^p - s} \right), \quad s = \text{sign} \left(\frac{\varepsilon_{32}}{\varepsilon_{21}} \right). \quad (3)$$

The term $q(p)$ vanishes only for a uniform refinement ratio ($r_{21} = r_{32}$). In the present mesh family $r_{21} = 1.742$ and $r_{32} = 1.314$, so $q(p)$ must be retained; since p appears on both sides of Eq. (2), the system is solved by fixed-point iteration. The extrapolated value and the fine-grid convergence index are then

$$f_{\text{ext}}^{21} = \frac{r_{21}^p f_1 - f_2}{r_{21}^p - 1}, \quad e_{21,a} = \left| \frac{f_1 - f_2}{f_1} \right|, \quad \text{GCI}_{\text{fine}} = \frac{1.25 e_{21,a}}{r_{21}^p - 1}. \quad (4)$$

5.2 Results by Mesh Level

5.2.1 Coarse Mesh — 370k cells

car_cd		

	Cd	

	car	0.29902237
car_cl		

	Cl	

	car	0.33571159

Figure 8: Coarse mesh: $C_d = 0.2990$, $C_l = +\mathbf{0.3357}$. Note the *positive* lift coefficient, consistent with the experimental value of $+0.345$ [3]. The medium and fine runs return a negative C_l of near-identical magnitude — see Section 7.2.

5.2.2 Medium Mesh — 837k cells

car_cd		

	Cd	

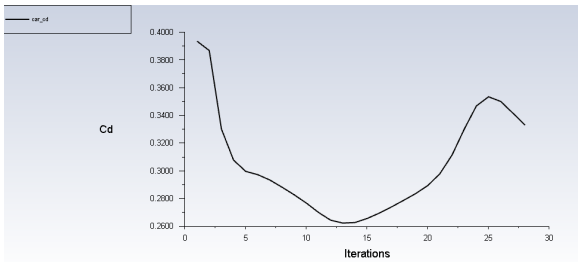
	car	0.28031334
car_cl		

	Cl	

	car	-0.32349919

Figure 9: Medium mesh: $C_d = 0.280$. The monitor shows $C_l = -0.323$ as originally reported; the force-vector sign error was identified subsequently and the corrected value is $+0.32349$ (Sec. 7.2). Moving-ground configuration.

5.2.3 Fine Mesh — 4.4M cells



(a) C_d convergence history (early iterations)

car_cd		

	Cd	

	car	0.2699273
car_cl		

	Cl	

	car	-0.32319065

(b) Final converged values

Figure 10: Fine mesh: $C_d = 0.270$. As with the medium mesh, the monitor shows the pre-correction $C_l = -0.323$; the corrected value is $+0.32319$ (Sec. 7.2). Moving-ground configuration — the stationary-ground rerun gives $C_d = 0.27908$, $C_l = +0.35525$ (Sec. 7.3).

5.3 GCI Summary

Table 4: Mesh independence summary. Apparent order $p = 3.53$ from the iterative solution of Eq. (2); asymptotic range check $\text{GCI}_{32}/(r_{21}^p \text{GCI}_{21}) = 0.96 \approx 1$.

Mesh	Cells	C_d	GCI (%)	Error vs. exp. (%)
Coarse	369 552	0.2990	—	+0.3
Medium	837 508	0.2800	5.25	−6.0
Fine	4 428 444	0.2699	0.77	−9.4
Richardson extrapolation	—	0.2682	—	−10.0
Meile et al. (2016)	—	0.298 ± 0.005	—	—

Stilt correction. The error column above compares a *stilt-free* CFD model against an experiment in which the body is mounted on four support stilts, and is therefore not a like-for-like comparison. Gutierrez et al. [6] decompose the drag by surface (their Table 3) and report a stilt contribution of $\Delta C_d = 0.0158$ (RSM) to 0.0245 (k - ω Standard). Subtracting this from the measured value gives an effective stilt-free reference of $C_d = 0.2735$ – 0.2822 , against which the moving-ground fine-mesh result of 0.2699 deviates by only -1.3% to -4.4% . This is the comparison the geometry actually supports. Correcting the ground condition as well (Sec. 7.3) raises the fine-mesh value to 0.27908 , placing it inside this band.

Because the apparent order $p = 3.53$ exceeds the formal second-order accuracy of the discretisation scheme, the GCI was recomputed with p fixed at the theoretical value of 2, as recommended by Celik et al. [7]. This yields $C_d^{\text{ext}} = 0.2649$ and $\text{GCI}_{\text{fine}} = 2.30\%$ — the same conclusion, with the fine-mesh numerical uncertainty remaining well below the deviation from experiment. The asymptotic range check is satisfied for $p = 3.53$ (0.96) but not for $p = 2$ (1.68), indicating the iteratively solved p is the more self-consistent estimate for this mesh family.

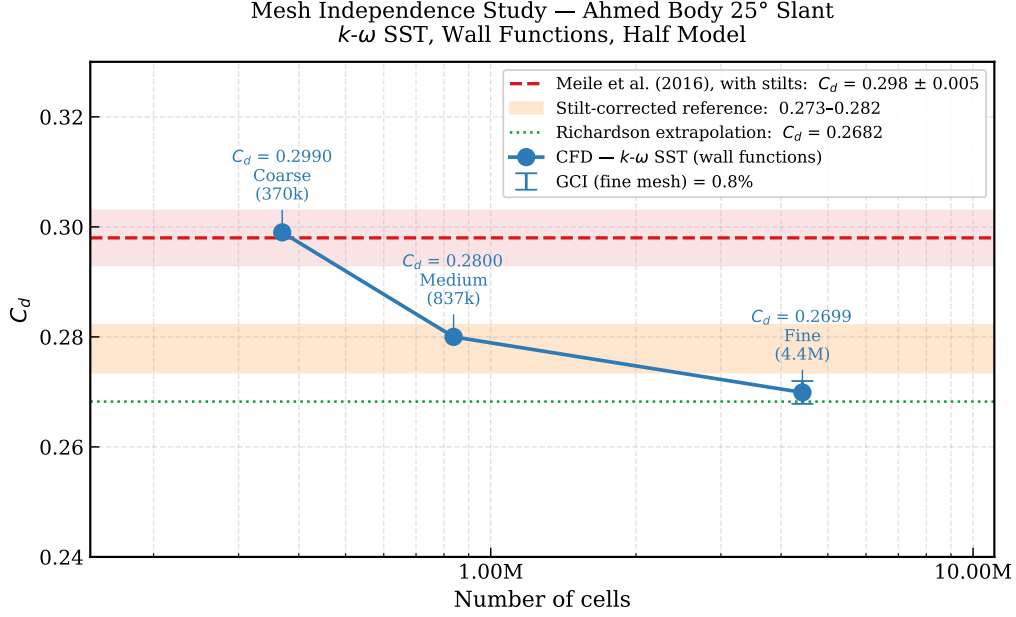


Figure 11: Mesh independence convergence. Red dashed: Meile et al. (2016) experimental C_d , measured on a stilt-mounted model. Orange band: stilt-corrected reference, the like-for-like target for this stilt-free geometry. Green dotted: Richardson-extrapolated C_d . Error bar: GCI uncertainty on the fine mesh.

6 Results and Validation

6.1 Aerodynamic Coefficients

Table 5: Fine-mesh aerodynamic coefficients vs. experimental reference, in the final configuration: stationary ground (matching the tunnels) with the lift force vector corrected.

Quantity	CFD (fine)	Experiment	Error
C_d (vs. stilt-corrected ref.)	0.27908	0.2735–0.2822	within range
C_d (raw, vs. stilted model)	0.27908	$0.298 \pm 0.5\%$ [4]	−6.4%
C_l	+0.35525	+0.345 [3]	+ 3.0%

Reaching this required two setup corrections, not physics tuning: the lift force vector (Sec. 7.2) and the ground boundary condition (Sec. 7.3). On the original moving-ground configuration the same mesh gave $C_d = 0.2699$ and $C_l = +0.32319$.

6.2 Experimental Wake Reference

The Lienhart & Becker wake measurements are available in unmodified form in the ERCOFTAC Classic Collection, Case 082 [5]. The dataset provides LDA profiles on downstream planes at $x = 0, 80, 200$ and 500 mm behind the body ($x/H = 0, 0.28, 0.69, 1.74$), with the origin at the rear face. These are reproduced in Figure 12 and are included in the repository ([data/ercoftac/](#)).

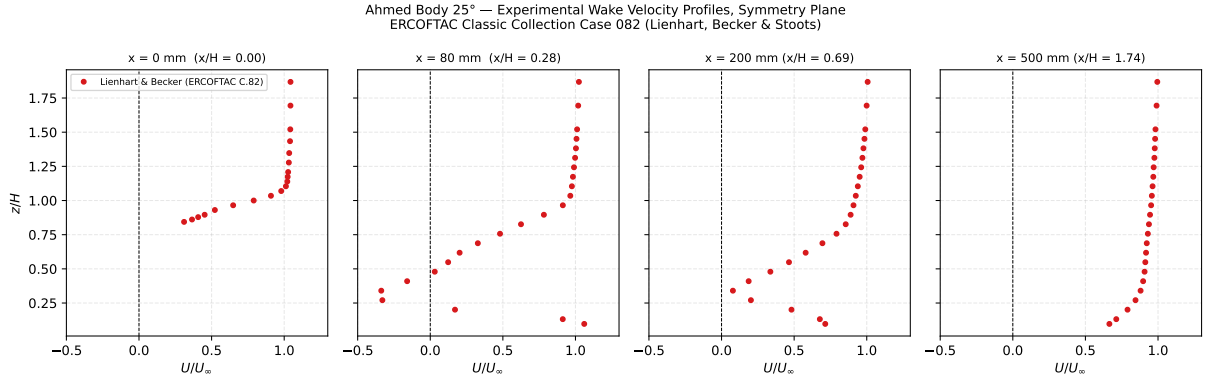


Figure 12: Experimental streamwise velocity profiles on the symmetry plane, 25° slant. Note the reverse flow at $x = 80$ mm ($U/U_\infty = -0.34$), the signature of the slant separation bubble. Data: ERCOFTAC Case 082 (Lienhart, Becker & Stoots).

A CFD-vs-experiment comparison against these profiles is **not presented in this revision**; see Section 7.5 for why.

7 Discussion

7.1 Drag Coefficient Underprediction

Against the stilt-corrected reference the fine mesh underpredicts C_d by 1.3–4.4%. The comparison against the raw measured value (−9.4%) overstates the error, because it charges the stilt-free CFD geometry with drag that the experimental model’s support stilts contributed.

With a fine-mesh GCI of only 0.77%, the *numerical* uncertainty is an order of magnitude smaller than the remaining deviation. The residual discrepancy is therefore **modelling error, not discretisation error**, and is consistent with the literature for $k-\omega$ SST with wall functions on bluff bodies:

- **Wall function buffer region:** $y^+ \approx 15$ –30 sits between the viscous sublayer and log-law validity, introducing near-wall velocity gradient errors.
- **RANS separation modelling:** $k-\omega$ SST underpredicts the slant recirculation zone at 25°, reducing base pressure drag.
- **Non-monotonic error vs. experiment:** Against the raw measured value the coarse mesh appears almost exact (+0.3%) while the fine mesh appears worst (−9.4%). This is an artefact of compensating errors: excess numerical diffusion on the coarse mesh inflates the predicted drag, which happens to offset both the model’s underprediction and the missing stilt drag. Agreement with experiment on an unconverged mesh is not evidence of a correct solution — the grid convergence study is what exposes this.

7.2 Lift Coefficient — Sign Error, Resolved

An earlier revision of this report carried an inverted C_l on the medium and fine meshes. The error has since been traced and corrected; it is documented here because the diagnosis is instructive.

The experimental value for the 25° slant is $C_l = +0.345$ [3] — *positive*. Gutierrez et al. [6] obtain +0.363 ($k\text{-}\varepsilon$ Realizable) and +0.376 ($k\text{-}\omega$ Standard), and explain the mechanism: at 25° the flow between the road and the underbody is slow, so the underbody static pressure remains higher than that on the upper surface, producing net positive lift. The medium and fine runs returned a negative value of the same magnitude.

That pattern was itself the diagnosis. A physical modelling deficiency cannot negate a coefficient while preserving its magnitude — it would corrupt the magnitude instead. The fault therefore had to lie in post-processing, and it did: the lift report definition’s force vector had been entered as $(0, -1, 0)$ on the medium and fine cases against $(0, 1, 0)$ on the coarse case, inverting the vertical axis. Correcting the vector and re-extracting C_l from the existing converged solutions — no re-iteration, so the flow fields are untouched — recovers the magnitudes to five significant figures:

Table 6: Lift coefficient across mesh levels, before and after correcting the force vector.

Mesh	C_l as-reported	C_l corrected	Deviation from exp.
Coarse (370k)	+0.3357	+0.3357 ²	−2.7%
Medium (837k)	−0.3235	+ 0.32349	−6.2%
Fine (4.4M)	−0.3232	+ 0.32319	−6.3%

All three meshes now agree on both sign and magnitude. The residual −6.3% is the same order as the C_d deviation.

However, this does not on its own constitute a validated C_l . The corrected value is a moving-ground result compared against a fixed-ground measurement, so the residual cannot yet be attributed to turbulence modelling — see Sec. 7.3.

7.3 Ground Boundary Condition — Mismatch Identified and Corrected

The ground condition originally simulated did not match either experiment it was compared against. The initial configuration applied a moving wall at $U = 40$ m/s, emulating a rolling road. Both experimental sources ran over a *stationary* floor:

Table 7: Ground condition in the reference experiments.

Source	Facility	Ground
Lienhart & Becker [2, 5]	LSTM, 3/4-open section	Fixed plate; model on 30 mm stilts
Meile et al. [3, 4]	Graz ISW	Fixed floor, six-component balance

The distinction is not cosmetic. Strachan et al. [8] measured this geometry over both a fixed and a moving ground plane and found that the moving plane thins the underbody boundary layer, increases the underbody flow speed, and displaces the near-wake vortex structure. Those are precisely the mechanisms that set the underbody static pressure — and therefore C_l , the coefficient already flagged in Sec. 7.2.

²The coarse case already carried the correct force vector; it is unchanged.

Corrected configuration

The fine mesh was therefore rerun with the ground boundary set to a stationary wall, restarted from the converged moving-ground solution with **no other setting altered**. The ground condition is thus the single isolated variable, and the comparison becomes like-for-like.

Convergence of the restart. Starting from an already converged field, the solution settled to residuals of order 10^{-6} within 70–80 iterations, satisfying the residual criterion of Table 3 with an order of margin. The force-monitor criterion in that table is, however, defined over a 200-iteration window, which a run this short does not span. The coefficients were steady over the iterations available, but the criterion *as written* has not been demonstrated across a full window; this is a reporting gap rather than evidence of a convergence failure, and closing it requires only a longer continuation run.

The point deserves care because Meile et al. [4] document a non-symmetric bi-stable wake mode for this geometry, and a change to the underbody flow is exactly the sort of perturbation that could excite a switch between states. It should be noted that the present half-model with an imposed symmetry plane *cannot* represent an asymmetric mode by construction: such a state is excluded by the boundary condition rather than resolved and rejected. This is a limitation of the half-model idealisation in its own right, independent of the restart length.

Table 8: Effect of the ground boundary condition on the fine-mesh force coefficients.

	Moving ground	Stationary ground	Experiment
C_l	+0.32319	+0.35525	+0.345 [3]
C_d	0.2699	0.27908	0.2735–0.2822 (stilt-corr.)
C_l deviation	−6.3%	+3.0%	—
C_d deviation	−1.3% to −4.4%	within range	—

Matching the ground condition improves both coefficients. C_l moves from −6.3% to +3.0% of the measured value — less than half the deviation, and now marginally above rather than below experiment. C_d rises into the stilt-corrected experimental band rather than sitting below it.

The relative magnitudes are themselves evidence for the mechanism. C_l changes by 9.9% against only 3.4% for C_d : lift is roughly three times more sensitive to the ground state than drag. This is precisely what Strachan et al. [8] would predict, since the underbody pressure field their moving plane perturbs acts principally on the *vertical* force.

An asymmetry in the two comparisons should be stated plainly. C_d is compared against a stilt-corrected reference, because the stilts contribute measurable drag that this stilt-free geometry cannot produce. No equivalent correction is applied to C_l , which is compared against the raw measured value. The justification is that the stilts are slender vertical cylinders: their *direct* contribution to the vertical force is small, whereas their contribution to the streamwise force is not. They do, however, disturb the underbody flow, so an indirect effect on lift cannot be excluded. Gutierrez et al. [6] resolve the drag contribution by surface but do not provide a correspondingly usable lift decomposition, so no correction is attempted here. The +3.0% figure therefore carries this residual uncertainty, and the C_l comparison is marginally less like-for-like than the C_d one.

The wider methodological point is that a substantial share of what could readily have been attributed to “ k - ω SST limitations” was in fact a boundary-condition mismatch. With the ground condition matched and the stilt geometry difference accounted for, the fine mesh reproduces the experiment to +3.0% in C_l and to within the reference band in C_d — without any change to the turbulence model or its coefficients.

Scope of the correction. Only the fine mesh was rerun on a stationary ground; the mesh independence study was performed on the moving-ground configuration. Since the GCI is a property of the spatial discretisation and no mesh parameter was altered, it is expected to carry over, but this has not been demonstrated and is left as a caveat rather than a claim.

7.4 Mesh Independence

The apparent order $p = 3.53$ exceeds the formal second-order accuracy of the discretisation scheme. Such super-convergence typically indicates that the mesh family is not perfectly self-similar: the aspect-ratio-based boundary layer sizing means the prism layers do not refine at the same rate as the poly-hexcore freestream cells, so y^+ is not held constant across mesh levels. The refinement ratios themselves are also non-uniform ($r_{21} = 1.742$ vs. $r_{32} = 1.314$), which is precisely why the $q(p)$ term in Eq. (3) cannot be neglected.

Reporting the GCI at both the iterated $p = 3.53$ (0.77%) and the theoretical $p = 2$ (2.30%) brackets the numerical uncertainty; in both cases it remains well below the 1.3–4.4% modelling error, so the conclusions are insensitive to the choice. A fixed first-layer thickness and a uniform refinement ratio would yield a cleaner mesh family and an apparent order closer to 2.

7.5 Wake Profiles — Withdrawn and Out of Scope

An earlier revision of this report presented a CFD-vs-experiment wake comparison with an MAE metric. That comparison has been **withdrawn rather than corrected**, because neither side of it was sound:

- **The experimental reference was not the experiment.** The profiles had been entered by hand at stations $x/H = 0.48, 0.61, 0.73, 0.86$. The ERCOFTAC Case 082 dataset [5] — the actual Lienhart & Becker measurements — provides planes at $x/H = 0, 0.28, 0.69, 1.74$. The stations that were plotted do not exist in the experiment, and the values did not match it: the real data shows reverse flow reaching $U/U_\infty = -0.34$ at $x = 80$ mm, an order of magnitude stronger than what had been tabulated.
- **The CFD profiles have no recorded station.** The Fluent export file `velocity_profiles.xy` labels its four blocks `wake_x1` through `wake_x4` and records no streamwise coordinate, so the location of the computed profiles cannot be recovered from the file.

Correcting the reference alone would not have made the comparison valid, since the computed side remains unlocated.

Wake validation is therefore outside the scope of this study, which validates the *force coefficients* against measurement. The two are distinct claims: C_d and C_l test

the integrated surface loads, whereas wake profiles test the resolved flow structure. Only the former is claimed here.

The experimental data is nonetheless included in the repository in unmodified form (`data/ercoftac/`), together with `scripts/ercoftac_wake_reference.py`, which reads and plots it. What a subsequent study would need to add is a re-export of the CFD symmetry-plane profiles at the four measured stations, with careful attention to the coordinate mapping: the ERCOFTAC origin lies at the *rear face* of the body, which does not coincide with the origin of the Fluent case. Mismapping those axes is what invalidated the withdrawn comparison in the first place.

8 Conclusions

- Mesh independence is achieved on the fine mesh: $GCI_{\text{fine}} = 0.77\%$ (2.30% if p is fixed at the theoretical second order), with the asymptotic range check satisfied (0.96). The Richardson-extrapolated value is $C_d = 0.2682$.
- In the final configuration — stationary ground, matching the experiments — the fine mesh gives $C_d = 0.27908$, inside the stilt-corrected experimental band (0.2735–0.2822), and $C_l = +0.35525$, within +3.0% of the measured +0.345. Since the numerical uncertainty is below 1%, these residuals are physical rather than a resolution artefact.
- Comparing a stilt-free geometry directly against the measured value inflates the apparent error to 9.4%. Likewise, the coarse mesh appears to agree almost exactly with experiment (+0.3%) purely through compensating errors. Both are reminders that a validation claim is only as good as the like-for-like nature of the comparison behind it.
- A lift sign error introduced between the coarse and medium runs has been traced to the force report definition’s direction vector, entered as $(0, -1, 0)$ instead of $(0, 1, 0)$, and corrected. Re-extracting C_l from the existing converged solutions gives +0.32349 (medium) and +0.32319 (fine), recovering the original magnitudes to five significant figures and bringing all three meshes into agreement on sign and magnitude (Sec. 7.2).
- The ground boundary condition had been specified as a moving wall at freestream speed, while both reference experiments used a stationary floor. Rerunning the fine mesh with a stationary ground — the only altered setting — moved C_l from -6.3% to +3.0% of the measured value and lifted C_d into the stilt-corrected band. Lift proved about three times more sensitive to the ground state than drag (9.9% against 3.4%), consistent with the underbody-pressure mechanism reported by Strachan et al. [8]. A material share of the deviation that might otherwise have been ascribed to turbulence modelling was therefore a boundary-condition mismatch (Sec. 7.3).
- The wake comparison presented in an earlier revision has been withdrawn: its experimental reference did not correspond to the actual measurements, and the computed profiles carry no recorded station coordinate (Sec. 7.5). Wake validation is accordingly outside the scope of this study, which validates the integrated force coefficients rather than the resolved wake structure. The genuine ERCOFTAC Case 082 data is included in the repository for any subsequent work.

References

- [1] Ahmed, S.R., Ramm, G., & Faltin, G. (1984). *Some salient features of the time-averaged ground vehicle wake*. SAE Technical Paper 840300.
- [2] Lienhart, H., & Becker, S. (2003). *Flow and turbulence structure in the wake of a simplified car model*. SAE Technical Paper 2003-01-0656. [*LDA wake velocity profiles; does not report force coefficients.*]
- [3] Meile, W., Brenn, G., Reppenhagen, A., & Fuchs, A. (2011). *Experiments and numerical simulations on the aerodynamics of the Ahmed body*. CFD Letters, 3(1), 32–39.
- [4] Meile, W., Ladinek, T., Brenn, G., Reppenhagen, A., & Fuchs, A. (2016). *Non-symmetric bi-stable flow around the Ahmed body*. International Journal of Heat and Fluid Flow, 57, 34–47.
- [5] ERCOFTAC Classic Collection, Case 082. *Flow around a simplified car body (Ahmed body)* — experiments by H. Lienhart, S. Becker & C. Stoots, LSTM Erlangen. <http://cfd.mace.manchester.ac.uk/ercoftac/doku.php?id=cases:case082>
- [6] Chavez Gutierrez, J.E., Vera Duarte, L.E., Oliveira Jr., A.A.M., & Cancino, L.R. (2020). *The Ahmed body’s external aerodynamics at 25° slant angle rear surface: a numerical analysis using CFD*. 18th Brazilian Congress of Thermal Sciences and Engineering (ENCIT 2020), ENC-2020-0060.
- [7] Celik, I.B. et al. (2008). *Procedure for estimation and reporting of uncertainty due to discretization in CFD applications*. Journal of Fluids Engineering, 130(7), 078001.
- [8] Strachan, R.K., Knowles, K., & Lawson, N.J. (2007). *The vortex structure behind an Ahmed reference model in the presence of a moving ground plane*. Experiments in Fluids, 42(5), 659–669.
- [9] Menter, F.R. (1994). *Two-equation eddy-viscosity turbulence models for engineering applications*. AIAA Journal, 32(8), 1598–1605.